

DPP-3

- Q.1 If $\vec{a} + \vec{b} + \vec{c} = 0$, $|\vec{a}| = 3$, $|\vec{b}| = 5$, $|\vec{c}| = 7$, then the angle between $\vec{a}$ & $\vec{b}$ is :
(A) $\pi/6$ (B) $2\pi/3$ (C) $5\pi/3$ (D) $\pi/3$
- Q.2 The lengths of the diagonals of a parallelogram constructed on the vectors $\vec{p} = 2\vec{a} + \vec{b}$ & $\vec{q} = \vec{a} - 2\vec{b}$, where $\vec{a}$ & $\vec{b}$ are unit vectors forming an angle of 60° are :
(A) 3 & 4 (B) $\sqrt{7}$ & $\sqrt{13}$ (C) $\sqrt{5}$ & $\sqrt{11}$ (D) none
- Q.3 Let $\vec{a}$, $\vec{b}$, $\vec{c}$ be vectors of length 3, 4, 5 respectively. Let $\vec{a}$ be perpendicular to $\vec{b} + \vec{c}$, $\vec{b}$ to $\vec{c} + \vec{a}$ & $\vec{c}$ to $\vec{a} + \vec{b}$. Then $|\vec{a} + \vec{b} + \vec{c}|$ is :
(A) $2\sqrt{5}$ (B) $2\sqrt{2}$ (C) $10\sqrt{5}$ (D) $5\sqrt{2}$
- Q.4 Given a parallelogram ABCD. If $|\vec{AB}| = a$, $|\vec{AD}| = b$ & $|\vec{AC}| = c$, then $\vec{DB} \cdot \vec{AB}$ has the value
(A) $\frac{3a^2 + b^2 - c^2}{2}$ (B) $\frac{a^2 + 3b^2 - c^2}{2}$ (C) $\frac{a^2 - b^2 + 3c^2}{2}$ (D) none
- Q.5 The set of values of x for which the angle between the vectors $\vec{a} = x\hat{i} - 3\hat{j} - \hat{k}$ and $\vec{b} = 2x\hat{i} + x\hat{j} - \hat{k}$ acute and the angle between the vector $\vec{b}$ and the axis of ordinates is obtuse, is
(A) $1 < x < 2$ (B) $x > 2$ (C) $x < 1$ (D) $x < 0$
- Q.6 If a vector $\vec{a}$ of magnitude 50 is collinear with vector $\vec{b} = 6\hat{i} - 8\hat{j} - \frac{15}{2}\hat{k}$ and makes an acute angle with positive z-axis then :
(A) $\vec{a} = 4\vec{b}$ (B) $\vec{a} = -4\vec{b}$ (C) $\vec{b} = 4\vec{a}$ (D) none
- Q.7 A, B, C & D are four points in a plane with pv's $\vec{a}$, $\vec{b}$, $\vec{c}$ & $\vec{d}$ respectively such that $(\vec{a} - \vec{d}) \cdot (\vec{b} - \vec{c}) = (\vec{b} - \vec{d}) \cdot (\vec{c} - \vec{a}) = 0$. Then for the triangle ABC, D is its
(A) incentre (B) circumcentre (C) orthocentre (D) centroid
- Q.8 Let $\vec{A}$ & $\vec{B}$ be two non parallel unit vectors in a plane. If $(\alpha\vec{A} + \vec{B})$ bisects the internal angle between $\vec{A}$ & $\vec{B}$, then α is equal to
(A) $1/2$ (B) 1 (C) 2 (D) -1
- Q.9 Image of the point P with position vector $7\hat{i} - \hat{j} + 2\hat{k}$ in the line whose vector equation is, $\vec{r} = 9\hat{i} + 5\hat{j} + 5\hat{k} + \lambda(\hat{i} + 3\hat{j} + 5\hat{k})$ has the position vector
(A) $(-9, 5, 2)$ (B) $(9, 5, -2)$ (C) $(9, -5, -2)$ (D) none
- Q.10 Let $\hat{a}$, $\hat{b}$, $\hat{c}$ are three unit vectors such that $\hat{a} + \hat{b} + \hat{c}$ is also a unit vector. If pairwise angles between $\hat{a}$, $\hat{b}$, $\hat{c}$ are θ_1 , θ_2 and θ_3 respectively then $\cos \theta_1 + \cos \theta_2 + \cos \theta_3$ equals
(A) 3 (B) -3 (C) 1 (D) -1

- Q.11 A tangent is drawn to the curve $y = \frac{8}{x^2}$ at a point $A(x_1, y_1)$, where $x_1 = 2$. The tangent cuts the x-axis at point B. Then the scalar product of the vectors $\vec{AB}$ & $\vec{OB}$ is
 (A) 3 (B) -3 (C) 6 (D) -6

- Q.12 L_1 and L_2 are two lines whose vector equations are

$$L_1 : \vec{r} = \lambda \left[(\cos \theta + \sqrt{3})\hat{i} + (\sqrt{2} \sin \theta)\hat{j} + (\cos \theta - \sqrt{3})\hat{k} \right]$$

$$L_2 : \vec{r} = \mu (a\hat{i} + b\hat{j} + c\hat{k}),$$

where λ and μ are scalars and α is the acute angle between L_1 and L_2 .

If the angle ' α ' is independent of θ then the value of ' α ' is

- (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$